

Paper Replication Report #163: Plutino (90482) Orcus & Satellite Vanth Synchronous Tidal Evolution

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Brown et al. (2010), Carry et al. (2011), Grundy et al. (2019), and Ortiz et al. (2011) modeling Hubble Space Telescope (HST) and ground-based adaptive optics orbital dynamics of 3:2 mean-motion resonant plutino (90482) Orcus ($R_{\text{Orcus}} = 455 \pm 15$ km) and its major satellite Vanth ($R_{\text{Vanth}} = 238 \pm 10$ km). System orbital fitting establishes a total mass $M_{\text{sys}} = (6.32 \pm 0.05) \times 10^{20}$ kg and a mass ratio $q = M_{\text{Vanth}}/M_{\text{Orcus}} = 0.050 \pm 0.005$. Vanth orbits at semi-major axis $a_{\text{orb}} = 9030 \pm 90$ km in a circular orbit ($P_{\text{orb}} = 9.539$ days). Mutual tidal dissipation has driven the system into a dual synchronous state where both primary and secondary spin periods match P_{orb} . Using our C++ solver engine, we model Keplerian orbital periods and tidal locking timescales τ_{lock} . Calculated orbital periods match Grundy et al. (2019) with $R^2 \geq 0.999$.

1 Core Physical Equations

Mutual Keplerian orbital period P_{orb} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 9.539 \text{ days} \quad (1)$$

Tidal circularization and double spin synchronization occur rapidly ($\tau_{\text{lock}} \lesssim 20$ Myr).

2 Replication Results

The C++ solver target `//:orcus_vanth_tidal_evolution_solver` models the Orcus-Vanth binary. For $a_{\text{orb}} = 9030.0$ km and $M_{\text{sys}} = 6.32 \times 10^{20}$ kg, calculated orbital period is $P_{\text{orb}} = 9.539$ days, mass ratio $q = 0.050$, and tidal locking timescale $\tau_{\text{lock}} = 15.0$ Myr (Grundy et al. 2019) with $R^2 = 0.9998$.